Definition 6.9

The variance of X is defined by

$$Var(X)=\sum\left[x-E(X)\right]^2 p(x) \quad \ldots (3)$$

if X is discrete random variable with probability mass function $p(x)$.

$$Var(X)=\int_{-\infty}^{\infty}\left[x-E(X)\right]^2 f_X(x)\,dx \quad \ldots (4)$$

if X is continuous random variable with probability density function $f_X(x)$.

Definition 6.10

Expected value of $[X-E(X)]^2$ is called the variance of the random variable.

i.e., $Var(X)=E\left[X-E(X)\right]^2=E\left(X^2\right)-\left[E(X)\right]^2 \quad \ldots (5)$

where

$$E\left(X^2\right)=\begin{cases}\sum_x x^2\,p(x), & \text{if } X \text{ is Discrete Random Variable}\\ \int_{-\infty}^{\infty} x^2 f(x)\,dx, & \text{if } X \text{ is Continuous Random Variable}\end{cases}$$

Note

- In the following examples, variance will be found using definition 6.10.
- The variances are defined only if the series in (3) is convergent or if the integrals in (4) exist.
- If X is a random variable, the standard deviation of X (S.D(X)), denoted by σ_X, is defined as $+\sqrt{Var[X]}$.
- The variance of X, denoted by σ_X^2 or $Var(X)$ or $V(X)$

DO YOU KNOW?

Mean is the center of gravity of a density; similarly, variance represents the moment of inertia of the same density with respect to a perpendicular axis through the center of gravity.

6.2.2 Properties of Mathematical expectation

(i) $E(a)=a$, where 'a' is a constant

(ii) $E(aX)=aE(X)$

(iii) $E(aX+b)=aE(X)+b$, where 'a' and 'b' are constants.

(iv) If $X\geq 0, then\ E(X)\geq 0$

(v) $V(a)=0$

(vi) If X is random variable, then $V(aX+b)=a^2V(X)$

Concept of moments

The moments (or raw moments) of a random variable or of a distribution are the expectations of the powers of the random variable which has the given distribution.

Definition 6.11

If X is a random variable, then the rth moment of X, usually denoted by φ'_r, is defined as $\varphi'_r =$

$$E\left(X^r\right)=\begin{cases}\sum_x x^r p(x), & \text{for discrete random variable}\\ \int_{-\infty}^{\infty} x^r f(x)dx, & \text{for continuous random variable}\end{cases}$$

provided the expectation exists.

Definition 6.12

If X is a random variable, the rth central moment of X about a is defined as $E\left[(X-a)^r\right]$. If $a=\varphi_x$, we have the rth central moment of X about φ_x, denoted by φ_r, which is

$$\varphi_r = E[(X-\varphi_X)^r]$$

Note

- $\varphi'_1 = E(X) = \varphi_X$, the mean of X.
- $\varphi_1 = E[X-\varphi_X] = 0$.
- $\varphi_2 = E[(X-\varphi_X)^2]$, the variance of X.
- All odd moments of X about φ_X are 0 if the density function of X is symmetrical about φ_X, provided such moments exist.

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